TechQuake: Visualizing Layoffs in the Technological Landscape

Introduction

Over the last three years, the United States has experienced significant employment growth. This general growth, however, stands in stark contrast to the situation within the technology sector, which has seen a notable downsizing trend. Despite the widespread consumer adoption of digital services and technologies, many tech companies have faced financial challenges leading to massive layoffs. Such a scenario raises concerns about the future of technological employment and the welfare of laid-off workers.

This issue is largely a domino effect of the pandemic. As COVID-19 restrictions eased, increased consumer spending led to the spike in inflation and interest rates. This shift prompted investors to favor companies with financially stable profits over aggressive growth. This realignment has resulted in massive layoffs across the tech industry.

Our project, *TechQuake*, aims to analyze the recent wave of layoffs in the tech industry from various perspectives. Through data visualization, we intend to provide insights into current economic trends, the stability of tech sectors, and the impacts on workforce dynamics.

Modality Decision

We truly believe in the value of being able to comprehensively understand the logic behind the unpredictable nature of tech layoffs. Such insights can ease various stakeholders in formulating more effective solutions to support the company and the laid-off victims.

Thus, we have decided to go with the **Design Studio** modality for our project. By pursuing a practical design and development approach, we can unpack and visualize hidden qualitative and quantitative patterns behind those layoffs. Ultimately, an interactive visualization dashboard makes for an impactful tool for our target audience.

Motivation and Goals

Data:

We will be using two different sources of datasets to design our interface. The *Layoffs.fyi* dataset contains post-COVID-19 layoff figures for tech companies worldwide. For this project, we will only focus on the 2000 businesses headquartered in the U.S. Our second dataset source is *Yahoo Finance*, where we will be grabbing financial data such as a company's market capitalization. We believe such information relates to the increasing worker layoffs, hence why we are also keen to visualize them.

Motivation and Aims:

In economic reports, it is common to see unemployment rates being visualized. These numbers comprise of individuals who never had jobs and total worker layoffs. The first problem has constantly been a popular topic of discourse. However, it is the other less-discussed cause of unemployment – the mass layoff of workers – that has become a critical issue post-COVID-19. Surprisingly, the technology sector happened to be the hardest hit industry, where even the major players are not spared from cutting jobs.

This deepening crisis has driven us to think about possible solutions that we could suggest through data visualization. We believe that **economic planners and organizations** require valuable insights into the nationwide tech layoff problem to develop effective mitigation strategies. Our main goal is to construct an interface to help them realize this mission. As guidance, we focus on these questions of interest:

1. How have tech layoffs evolved over time? Can they be correlated with major economic or industry-specific events?
2. Which subsector within the tech industry is most susceptible to workforce adjustments?
3. Do the financing stages of a tech company impact layoffs?
4. How concentrated are the tech layoffs in the U.S.?

Our design will enable economic planners and related organizations to interact with different measures, both on the company-level and country-level. We plan to utilize historical data to graph time-series plots of tech layoffs. With this, our clients can have a solid understanding of the general trend and the ability to make various comparisons across companies. Additionally, we aim to include a map of the U.S. to visualize the geographical distribution of tech layoffs, allowing our clients to identify regional patterns and the tech hubs with most job cuts.

We also intend to integrate graphs relating to the relevant companies' financial performance, such as market capitalization and corporate taxes. This enables our targeted clients to make a more in-depth analysis and draw connections between tech layoffs and key financial indicators.

By including the ability to visualize temporal, geographical, and sector/company data, we feel that our interactive dashboard design is all-inclusive. Hence, we are confident that the economic planners and relevant organizations' needs of valuable tech layoff insights will be satisfied.

Literature Review

As mentioned in the previous section, there has been a lack of prior visualization work pertaining to worker layoffs, let alone within the tech industry. Despite this, there are an abundance of used concepts that can be applied to visualize tech layoffs. One of the closest prior works was the health economic models by Smith (2020). This was done using Shiny as it allows users to interact and visualize the data without changing any code. Furthermore, he emphasized on a flexible and user-friendly interface to communicate his intended results. This relates to our goal, which is to ease our clients in navigating through the multi-layered tech layoff data.

Additionally, we want to find the geographical relations of tech layoff in the U.S. Liu (2016) successfully portrayed the gross domestic product (GDP) relationship between states through the integration of a GDP-based map. Applying a similar concept to our project enables the economic planner to see the layoff data from a geographical standpoint, thus adding an extra layer of interactivity to our dashboard.

To convey more information in visualization, it is better to use faceting (under the ggplot2 package) instead of overloading them in a single graph (Schwabish, 2014, p. 212). As we have learned in class, faceting makes it easier to see trends in different categories. This can be applied in our project, specifically when comparing layoff data between different tech companies. Another approach in making our visualizations more effective is to construct boxplots, which offers a succinct overview of the distribution of the dataset and makes it possible to spot outliers (Suvarnapathaki, 2023). This would be beneficial when comparing different subsectors/ financing stages of a tech company and layoffs.

Some of the challenges that we might encounter is that economic planners usually work with static graphs (Schwabish, 2014, p. 228). However, we believe that by focusing on interactivity, it will benefit our clients. This is because interactive visualizations "allows a transfer of information between the figure and the user" (Schwabish, 2014, p. 227) which is crucial for our clients to better understand tech layoffs. In turn, economic planners and organizations are able to make more informed strategies to resolve such crisis.

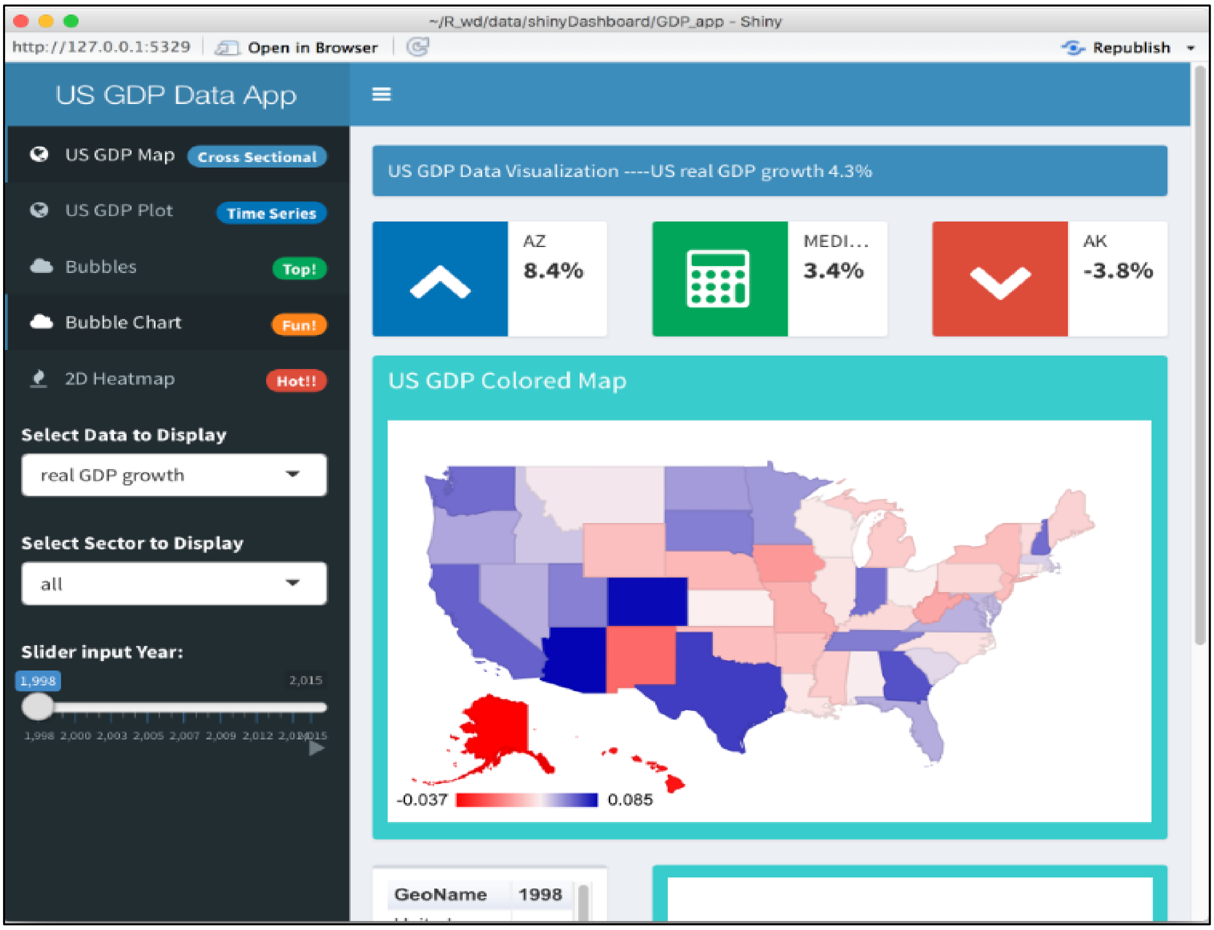

Figure 1: Map-based dashboard using Shiny.

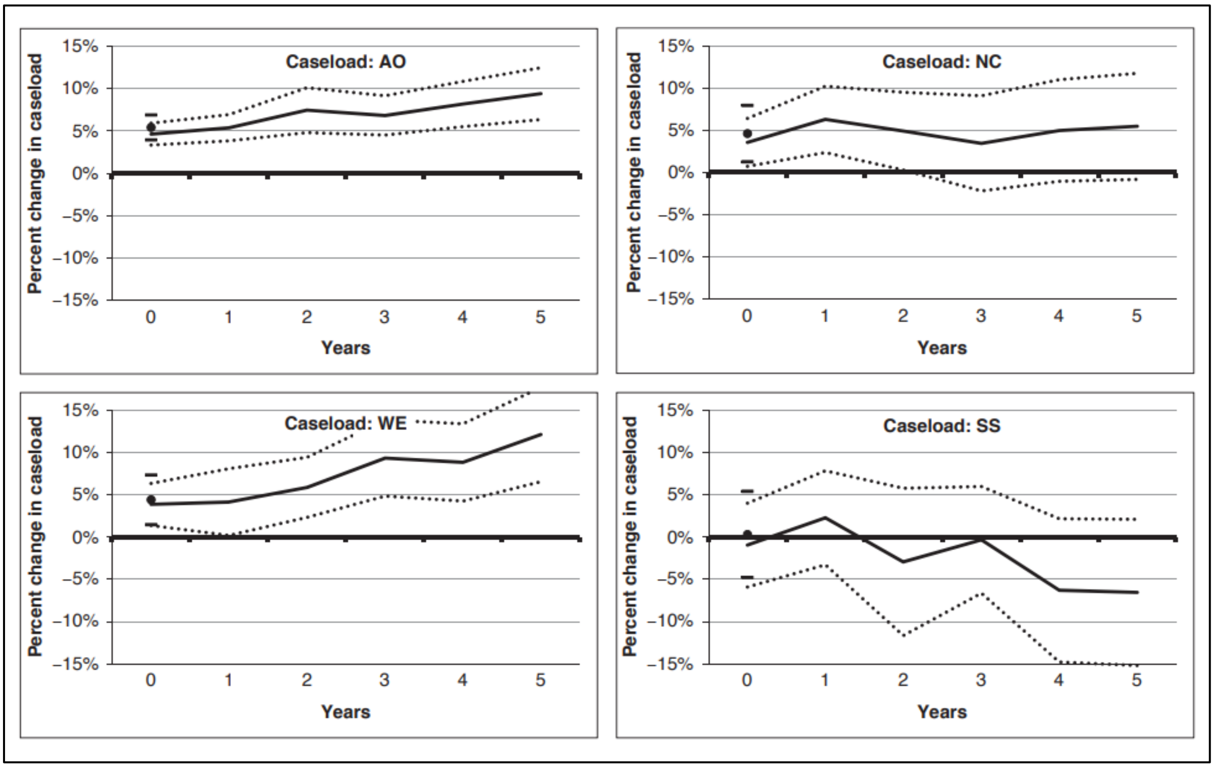

Figure 2: Use of faceting to compare different cases.

References

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